CIV E 779 Machine Learning for Engineers - Assignment 4

Instructor: Qipei (Gavin) Mei (qipei.mei@ualberta.ca); Yuezhen Gao (yuezhen@ualberta.ca)

Student Name:

CCID:

Question 1 [30 Marks]

Concrete is a composite material composed of fine and coarse aggregate bonded together with a fluid cement (cement paste) that hardens (cures) over time. It is the most widely used building material. The compressive strength of concrete is a very important property because it determines whether a given concrete mixture will meet the needs of a specific project. The compressive strength of a concrete mixture is dependent on many factors, such as material components and curing time (age), and therefore is very hard to predict.

In this question, you are given a dataset of experimental results, including these factors and the measured compressive strength. You are asked to build a neural network model to predict the compressive strength of concrete based on the input information. To answer this question, you should write a report with proper tables and figures summarizing the work you do for this question. The following requirements should be met:

1. You should build a neural network model. You are allowed to use any python packages, such as sklearn to build the models.
2. You should check and avoid the bias/variance issues of the model.
3. You should use k-fold cross-validation to evaluate the performance of the models.
4. You should choose a single metric to evaluate the performance of your model and provide justification.
5. You should show your effort to improve the performance of the model as much as you can. You can try different strategies, such as coming up with different input features, using one-hot encoding, different learning rates, etc., to improve the performance of your model (refer to the lecture notes to find more ways to improve your model.
6. You should report the performance of the models on the test set.

Dataset:
The dataset is given as two csv files together with the assignment. The dataset has been split into two sets (train/val and test). You can further split the data for validation purposes. The description of each column is presented below:

- Cement (component 1) -- quantitative -- kg in a m^3 mixture -- Input Variable
- Blast Furnace Slag (component 2) -- quantitative -- kg in a m^3 mixture -- Input Variable
- Fly Ash (component 3) -- quantitative -- kg in a m^3 mixture -- Input Variable
- Water (component 4) -- quantitative -- kg in a m^3 mixture -- Input Variable
- Superplasticizer (component 5) -- quantitative -- kg in a m^3 mixture -- Input Variable
- Coarse Aggregate (component 6) -- quantitative -- kg in a m^3 mixture -- Input Variable
- Fine aggregate (component 7) -- quantitative -- kg in a m^3 mixture -- Input Variable

- Age -- quantitative -- Day (1~365) -- Input Variable
- Concrete compressive strength -- quantitative -- MPa -- Output Variable

For your reference, below is the paper that released the original dataset:
Yeh, I.C., 1998. Modeling of strength of high-performance concrete using artificial neural networks. Cement and Concrete Research, 28(12), pp.1797-1808.

Question 2 [10 Marks]

This question involves modifying the "CIV_E_779_Lecture_8_CNN_Example.ipynb" code, which is an example of using a convolutional neural network for handwritten digit recognition. The task is to complete step 9 of the code, which involves testing your own image on the model. To properly train the model, you can either set up the coding environment locally if you have a GPU available, or use a cloud-based platform like Google Colab (https://colab.research.google.com).

To answer this question, you will need to demonstrate your work to complete step 9 and discuss the results. Note that while the provided code includes an example of handwritten digit recognition using a specific image, you may need to process your image differently so that it matches the format of the training/testing images.

Questions 3: Quiz [10 Marks]

Note there is only one correct answer for each question.

Please provide your answers **in your submitted report.**

1. What does the process of 'forward propagation' in neural networks involve?

A) Adjusting weights based on errors

B) Calculating cost functions

C) Passing input data through the network layers to get output

D) Normalizing data before inputting to the network

2. What is 'backpropagation' used for in neural networks?

A) Data normalization

B) Error correction and weight adjustment

C) Feature extraction

D) Data visualization

3. What happens if all weights in a neural network are initialized to zero?

A) The network will learn effectively

B) The gradients will be different for each neuron

C) The network will get stuck due to symmetry

D) The network will converge faster

4. What is the main purpose of gradient checking?

A) To ensure the forward propagation code is correct

B) To verify the correctness of backpropagation

C) To optimize the learning rate

D) To adjust the weights in the network

5. What is the typical output of a multi-class classification neural network?

A) Single output unit

B) Multiple output units representing each class

C) Continuous regression values

D) Confusion matrix

6. How is the gradient of the cost function computed during backpropagation?

A) By numerical estimation of derivatives

B) By calculating the derivatives of the cost function with respect to each weight

C) By averaging the input data

D) By reducing the number of hidden layers

7. Which of the following best describes deep learning?

A) A machine learning method based on decision trees

B) A subset of neural networks that use a large number of layers for representation learning

C) A statistical approach to linear regression models

D) An unsupervised learning technique for clustering

8. What is the effect of using max pooling in a CNN?

A) Reducing the number of channels in the image

B) Subsampling the image to reduce its size

C) Increasing the resolution of the image

D) Adding additional layers to the network

9. Which of the following is true about padding in CNNs?

A) It increases the size of the output feature map.

B) It reduces the computation time.

C) 'Same' padding keeps the output size the same as the input size.

D) 'Valid' padding adds extra pixels to the input.

10. If we have an input image of size 7x7, a kernel of size 3x3, a stride of 1, and no padding, what will be the output size?

A) 6x6

B) 5x5

C) 7x7

D) 4x4